

Design Off Grid Solar System



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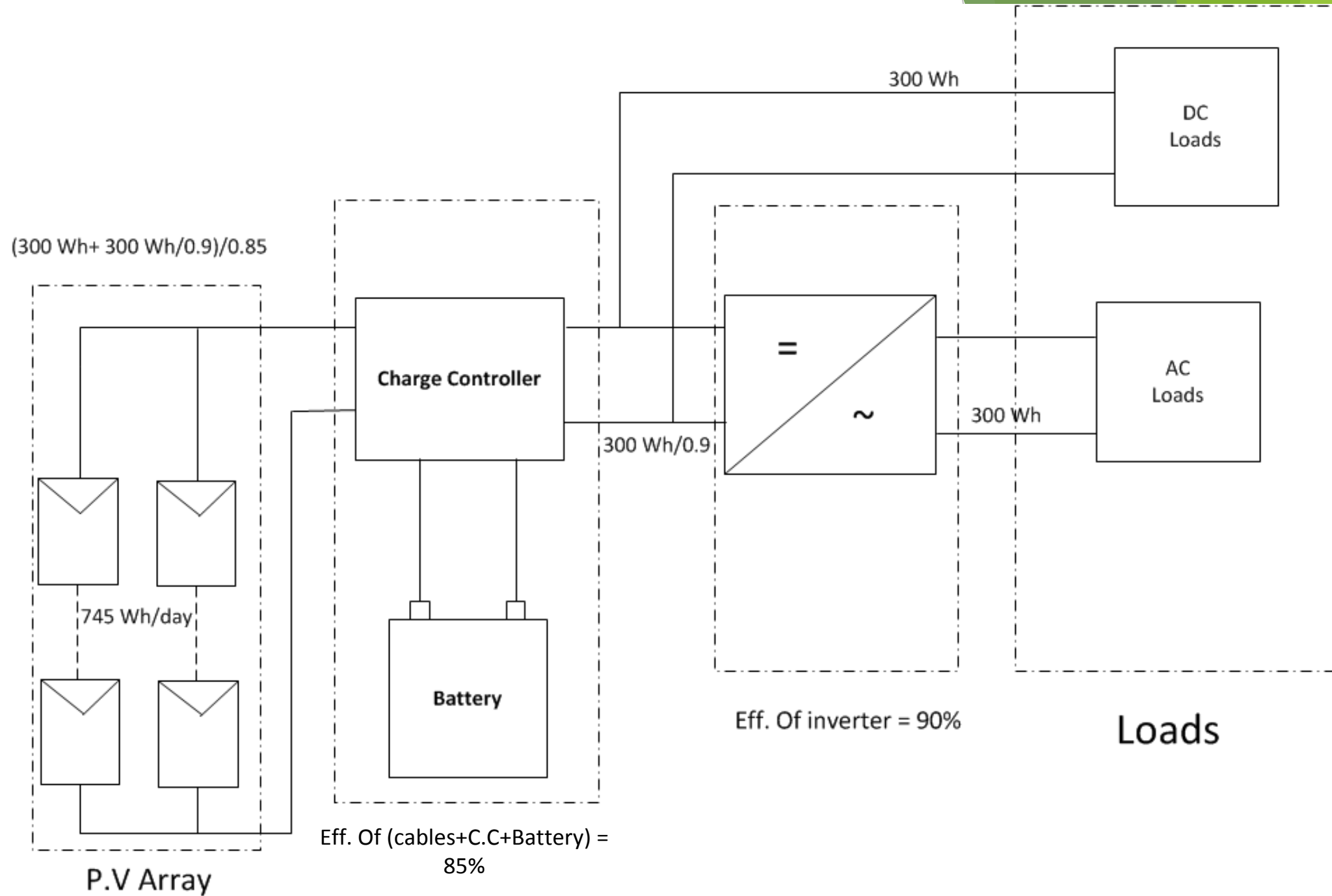
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Let's take an example :

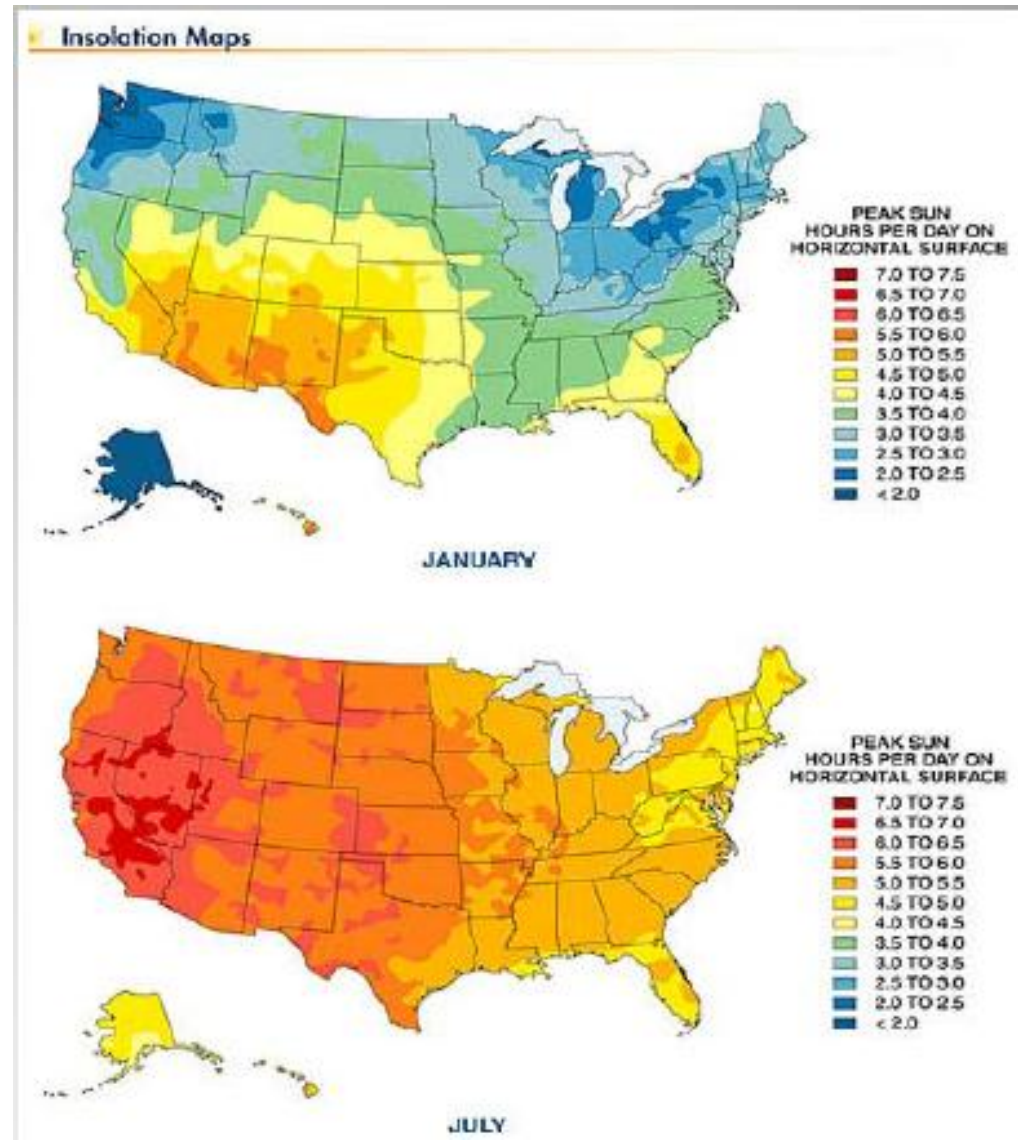
Home has :

	AC		DC	
	Tv	Computer	Lighting Unit (DC)	The Unit
Qty	1	1	4	unit
power/Unit	100	100	25	watt
Total Power	100	100	100	watt
Time of Use	2	1	3	hr/day
Total Energy	200	100	300	wh/day
1	total energy for Ac =300 wh		total energy for DC =300 wh	

2) Losses



3) Equavalent sun hours = 4.5 hr / day



4) P.V Arrays

-Peak power = $745 / 4.5 = 165.6$ watt

-Number of Panels = $165.6 / 100 = 1.7$

So Number of Panel = **2**

P.V Configuration

Parallel	Series
$I (\text{max}) = 7 * 2 = 14$ A	$V (\text{max}) = 20 * 2 = 40$ v

Panel Specs

Peak power (w)	100
Vmpp (v)	16
Impp (A)	6.25
Vo.c (v)	20
Is.c (A)	7

5) Charge controller design

When P.V is Parallel	When P.V is Series
V max (C.C) > V max (P.V) True	V max (C.C) > V max (P.V) True
I max (C.C) > I max (P.V) False	I max (C.C) > I max (P.V) True

Charge Controller Specs

V max	60
I max	10
Operational Voltage	12v / 24v
MPPT	Yes

So , Panels Should be in Series

6) Battery Design

$$\begin{aligned}\text{Total Capacity of Battery} &= \frac{\text{Total energy demand} * \text{auto. Days}}{\text{D.O.D} * \text{Operational Voltage} * \text{efficiency}} \\ &= 103.5 \text{ Ah}\end{aligned}$$

Battery in series	Battery in Parallel
$= 24 / 12 = 2$ batteries	$= 103.5 / 21 = 5$ batteries

Battery Specs	
D.O.D	60 %
Battery Voltage (v)	12
Battery Capacity (AH)	21

So , Number Of Batteries = $2 * 5 = 10$

7) Inverter Design

Where ,

$$\text{Eff.} = \text{Output} / \text{Input}$$

So, Min Power Rating = 200 w / 0.9

AC Loads only

= 222.2

watt

Inverter Specs

Efficiency	90 %
Operational Voltage	24 v

Cables sizing :

$$S(mm^2) = \frac{0.3 * L(m) * I_{Max}(A)}{\Delta V(\%)}$$

Where :

- **L** >>> cable length .
- **I max** >>> max current of cable = 1.56 * I (load) .
- **ΔV** >>> 1 % to 5 %

Notes :

- **ΔV** —————> Not excesses 3 % between Panels and charge controller
- **ΔV** —————> Not excesses 1 % between battery and charge control
- **ΔV** —————> Not excesses 5 % between charge controller and Load

Layout :

So our system has :
1 string

- String gives 6.25 A
- The operational voltage = 24 V
- So we need
combiner box greater than 6.25 A

